

LOSO Experiment B_flex_ypr_lstm

LSTM only · Flex + YPR (both hands), no Nothing

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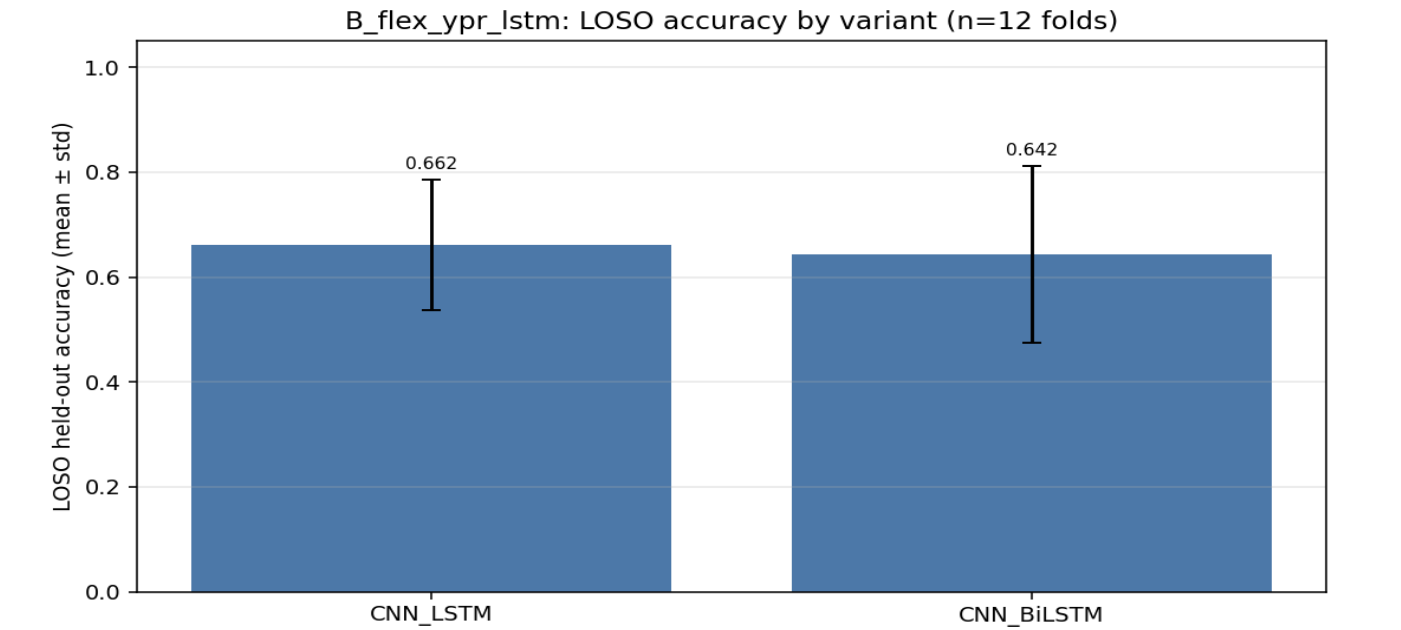
Experiment configuration

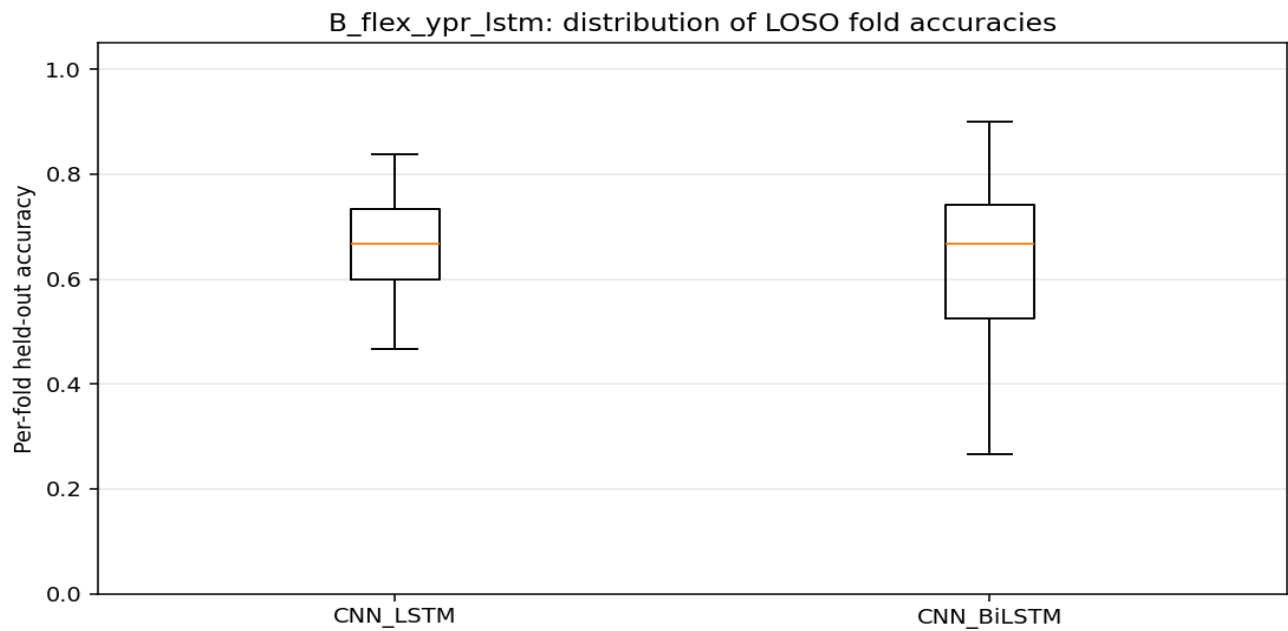
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, YPR
Hands	left, right
Sensor channels	98
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

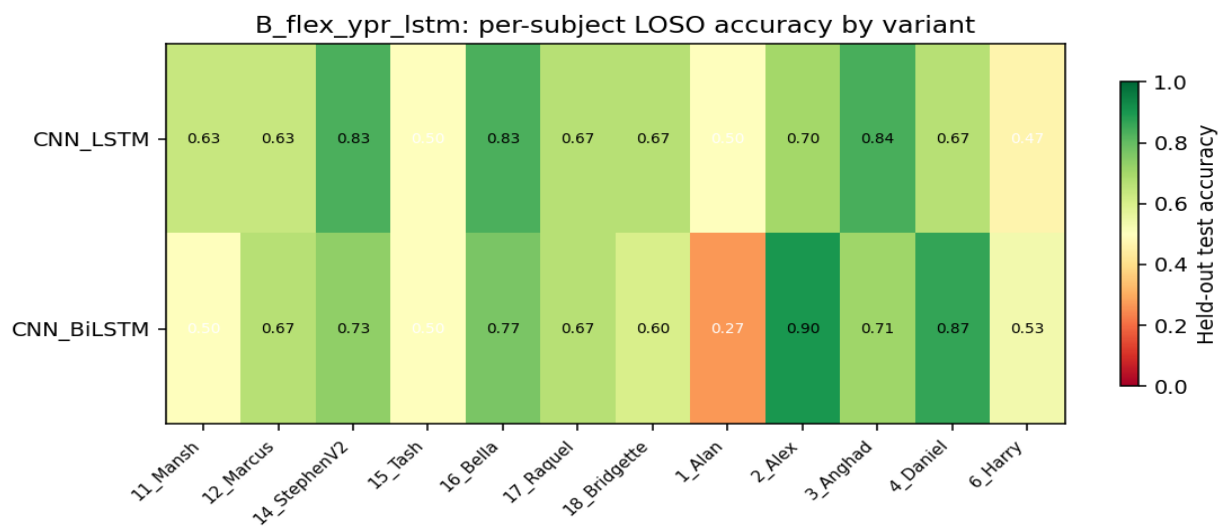
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	58,406	0.6616	0.1237	0.4667	0.8387	0.7260
CNN_BiLSTM	95,526	0.6425	0.1687	0.2667	0.9000	0.7688

Best variant by LOSO mean: **CNN_LSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	0.633	0.500
12_Marcus	331	0.633	0.667
14_StephenV2	331	0.833	0.733
15_Tash	331	0.500	0.500
16_Bella	331	0.833	0.767
17_Raquel	331	0.667	0.667
18_Bridgette	331	0.667	0.600
1_Alan	331	0.500	0.267
2_Alex	331	0.700	0.900
3_Anghad	330	0.839	0.710
4_Daniel	331	0.667	0.867
6_Harry	331	0.467	0.533

Variant: CNN_LSTM

Params: 58,406

LOSO mean accuracy: 0.6616 ± 0.1237 (min 0.4667, max 0.8387)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	0/5	5/5	5/5	4/5	5/5	0/5	0.633
12_Marcus	5/5	5/5	5/5	0/5	4/5	0/5	0.633
14_StephenV2	5/5	5/5	5/5	0/5	5/5	5/5	0.833
15_Tash	3/5	5/5	5/5	0/5	2/5	0/5	0.500
16_Bella	4/5	5/5	3/5	4/5	5/5	4/5	0.833
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
18_Bridgette	5/5	5/5	0/5	2/5	5/5	3/5	0.667
1_Alan	5/5	5/5	5/5	0/5	0/5	0/5	0.500
2_Alex	5/5	4/5	5/5	0/5	4/5	3/5	0.700
3_Anghad	5/5	5/5	5/5	0/5	6/6	5/5	0.839
4_Daniel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
6_Harry	5/5	2/5	5/5	0/5	1/5	1/5	0.467
MEAN	0.867	0.933	0.883	0.167	0.767	0.350	0.662

Variant: CNN_BiLSTM

Params: 95,526

LOSO mean accuracy: 0.6425 ± 0.1687 (min 0.2667, max 0.9000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	5/5	0/5	5/5	0/5	0.500
12_Marcus	5/5	5/5	5/5	0/5	5/5	0/5	0.667
14_StephenV2	2/5	5/5	5/5	0/5	5/5	5/5	0.733
15_Tash	3/5	5/5	5/5	2/5	0/5	0/5	0.500
16_Bella	3/5	5/5	5/5	0/5	5/5	5/5	0.767
17_Raquel	5/5	5/5	5/5	0/5	5/5	0/5	0.667
18_Bridgette	0/5	5/5	0/5	5/5	3/5	5/5	0.600
1_Alan	5/5	0/5	3/5	0/5	0/5	0/5	0.267
2_Alex	5/5	3/5	5/5	5/5	5/5	4/5	0.900
3_Anghad	5/5	5/5	5/5	0/5	6/6	1/5	0.710
4_Daniel	5/5	5/5	5/5	1/5	5/5	5/5	0.867
6_Harry	5/5	2/5	5/5	0/5	4/5	0/5	0.533
MEAN	0.800	0.750	0.883	0.217	0.783	0.417	0.642